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Question Paper Code : 16371

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2020

Third Semester

B.E. – Computer Science and Engineering

19CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

A, CO1 1. Convert the two decimal numbers 6 and 14 to 5-bit 2's-complement numbers, and then perform addition on them. Indicate whether or not overflow occurs or not.

A, CO1 2. Consider registers R_2 and R_5 contain the decimal numbers 2140 and 3070 respectively before each of the following addressing modes is used to access a memory operand. The effective address (EA) in each case for the given below are _____ and _____ . Justify your answer.

A) 12(R_3) B) 65(R_2, R_5)

A, CO2 3. Represent the following floating point number in IEEE single-precision format:

$-101.11011 \times 2^{-42}$

A, CO2 4. In an n × n array multiplier, the critical path has a total of _____ gate delays, including the initial AND gate delay in all cells. Justify your answer.

A) 6(n − 1) + 1 B) 6(n − 1) C) 6(n − 1) – 1 D) 6(n + 1) − 1

U, CO3 5. In hardwired control, write down the logic expression for "Counter_enable" control signal while dealing with memory delays. Why is it necessary?

U, CO3 6. In a pipelined system, the periods of time when the unit is idle is called as _____. Justify.

A) stalls B) bubbles C) hazards D) both (A) and (B)

U, CO4 7. Define DDR-SDRAM and its significance.

A, CO4 8. A block-set-associative cache consists of a total of 64 blocks, divided into 4-block sets. The main memory contains 4096 blocks, each consisting of 32 words. Assuming a 32-bit byte-addressable address space, how many bits are there in each of the tag, set, and word fields?

A) 22, 4, 6 B) 21, 4, 7 C) 20, 4, 8 D) 21, 5, 6

U, CO5 9. The following are some events that occur after a device controller issues an interrupt while process L is under execution.

(P) The processor pushes the process status of L onto the control stack.

(Q) The processor finishes the execution of the current instruction.

(R) The processor executes the interrupt service routine.

(S) The processor pops the process status of L from the control stack.

(T) The processor loads the new PC value based on the interrupt.

Which one of the following is the correct order in which the events above occur? Justify.

A) QPTRS B) PTRSQ C) TRPQS D) QTPRS

U, CO5 10. Compare and contrast between memory-mapped I/O and I/O mapped I/O.

PART B – (5 × 16 = 80 Marks)

U, CO1 11. a) i. "Performance of the system can be increased by performing a number of operations in parallel". Brief on the different levels by which parallelism can be implemented. (6 Marks)

U, CO1 11. a) ii. Highlight the various addressing modes with suitable example. (10 Marks)

OR

U, CO1 11. b) i. With a neat sketch, brief on the connection between the processor and the main memory. (6 Marks)

U, CO1 11. b) ii. Explain the concept of instruction execution and straight-line sequencing by considering with and without conditional branching instruction. (10 Marks)

U, CO2 12. a) i. Explain the working of 4-bit carry-lookahead adder with a neat diagram. (8 Marks)

A, CO2 12. a) ii. Multiply the following pair of signed 2's-complement numbers using the Booth algorithm and explain. Assume, A is the multiplicand and B is the multiplier.

A = 1110011 and B = 1010100 (8 Marks)

OR

U, CO2 12. b) i. With a neat diagram, explain fast multiplication methods in detail. (8 Marks)

A, CO2 12. b) ii. Explain restoring division technique with a neat diagram and using the same perform 23%5. (8 Marks)

U, CO3	13. a) i.	Identify the various control signals involved in the datapath of a RISC-style processor, with a neat diagram.	(8 Marks)
U, CO3	13. a) ii.	In a pipeline system, give suitable example for data dependency between these two instructions and show how stalling can be avoided.	(8 Marks)

OR

U, CO3	13. b) i.	With a neat diagram, explain the three-bus CISC-style processor organization. Give the sequence of steps needed to fetch and execute the instruction: Load R3, (R5)+.	(8 Marks)
U, CO3	13. b) ii.	Describe different methods for branch prediction in a pipeline system and explain.	(8 Marks)
A, CO4	14. a) i.	Describe how to form a large memory structure for a 16 M × 32 memory using 1 M × 4 memory chips with a suitable diagram.	(8 Marks)
U, CO4	14. a) ii.	Explain the working principle of direct memory access (DMA).	(8 Marks)

OR

A, CO4	14. b) i.	A computer has an 8 GByte memory with 64-bit word sizes. Each block of memory stores 16 words. The computer has a direct-mapped cache of 128 blocks. The computer uses word level addressing. What is the address format? Explain. If the cache is changed to a 4-way set associative cache, what is the new address format? Explain.	(8 Marks)
U, CO4	14. b) ii.	With suitable diagrams, explain the working of static and dynamic RAM cell.	(8 Marks)
U, CO5	15. a) i.	Explain with a neat diagram, the working of parallel input interface.	(8 Marks)
U, CO5	15. a) ii.	Consider the situation where a number of devices capable of initiating interrupts are connected to the processor. Assume, several devices may request interrupts at exactly the same time. How to handle the situation?	(8 Marks)

OR

U, CO5	15. b) i.	Explain with a neat diagram, the implementation of polling using device register.	(8 Marks)
U, CO5	15. b) ii.	Explain the USB device's characteristics and architecture.	(8 Marks)